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Database Programming with PL/SQL

9-5

Review of Object Privileges

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Objectives

- This lesson covers the following objectives:
 - List and explain several object privileges
 - Explain the function of the EXECUTE object privilege
 - Write SQL statements to grant and revoke object privileges

Purpose

- You already know that one of the benefits of PL/SQL subprograms is that they can be reused in many applications
- Users can call and execute subprograms only if they have the privileges to do so
- This lesson first reviews object privileges in general, then focuses in more detail on the privileges needed to execute a PL/SQL subprogram

What Is an Object Privilege?

- An object privilege allows the use of a specific database object, such as a table, a view, or a PL/SQL procedure, by one or more database users
- When a database object is first created, only its owner (creator) and the Database Administrator are privileged to use it
- Privileges for all other users must be specifically granted (and maybe later revoked)
- This can be done by the object's owner or by the DBA

To grant an object privilege, you must be the owner of the object, have been granted object privileges with the GRANT OPTION by the object owner, or have been granted the GRANT ANY OBJECT PRIVILEGE system privilege. DBA's can grant privileges on and access any object in the database.

What Object Privileges Are Available?

- Each object has a particular set of grantable privileges
- The following table lists the privileges for various objects

Object Privilege	Table	View	Sequence	Procedure
ALTER	X		X	
DELETE	X	X		
EXECUTE				X
INDEX	X			
INSERT	X	X		
REFERENCES	X	X		
SELECT	X	X	X	
UPDATE	X	X		

What Object Privileges Are Available?

- SELECT, INSERT, UPDATE, and DELETE privileges allow the holder (the grantee) of the privilege to use the corresponding SQL statement on the object
- For example, INSERT privilege on the EMPLOYEES table allows the holder to INSERT rows into the table, but not to UPDATE or DELETE rows



What Object Privileges Are Available?

- The ALTER privilege allows the grantee to ALTER the table, while INDEX privilege allows the grantee to create indexes on the table
- Of course, you can automatically do this on your own tables
- The REFERENCES privilege allows the grantee to check for the existence of rows in a table or view using foreign key constraints



Why do we need REFERENCES privilege?

The REFERENCES privilege allows a user's session to read data from a table, but ONLY in order to CHECK foreign key constraints.

Granting Object Privileges

Syntax:

```
GRANT object_priv [(columns)]  
ON object  
TO {user|role|PUBLIC}  
[WITH GRANT OPTION];
```

Examples:

```
GRANT INSERT, UPDATE  
ON employees TO TOM, SUSAN;  
  
GRANT SELECT  
ON departments TO PUBLIC;
```

Syntax	Definition
object_priv	The privilege to be granted.
columns	Limits UPDATE privilege to a specific column (optional).
object	The object name on which the privilege(s) are granted.
user role	Identifies the user/roles to whom the privilege is granted.
PUBLIC	Grants the named privilege(s) to all users.
WITH GRANT OPTION	Allows grantee to grant object privileges to others.

To invoke a subprogram, a user needs only EXECUTE privilege on the subprogram. He/she MAY NOT need any privileges on the objects referenced by SQL statements within the subprogram, but that depends on the code in that stored subprogram, and how security is handled in that program.

Remember:

We can GRANT (or REVOKE) several privileges to/from several users in a single command by using comma-separated lists, as shown in the first example.

Any privilege granted to PUBLIC is automatically granted to all users.

WITH GRANT OPTION allows the grantee to grant (pass on) the privilege to other users.

Only the UPDATE privilege can reference individual columns.

Revoking Object Privileges

- Syntax:

```
REVOKE object_priv [(columns)]  
ON object  
FROM {user|role|PUBLIC};
```

- Examples:

```
REVOKE INSERT, UPDATE ON employees FROM TOM, SUSAN;  
  
REVOKE SELECT ON departments FROM PUBLIC;
```



Using the EXECUTE Privilege With Stored Subprograms

- To invoke and execute a PL/SQL subprogram, the user must be granted EXECUTE privilege on the subprogram

```
CREATE OR REPLACE PROCEDURE add_dept ... ;  
CREATE OR REPLACE FUNCTION get_sal ... ;  
  
GRANT EXECUTE ON add_dept TO TOM, SUSAN;  
GRANT EXECUTE ON get_sal TO PUBLIC;  
...  
REVOKE EXECUTE ON get_sal FROM PUBLIC;
```



Referencing Objects in Subprograms

- What about the objects referenced inside the subprogram?
- To invoke a subprogram, a user needs only EXECUTE privilege on the subprogram

```
CREATE OR REPLACE PROCEDURE add_dept ...  
IS BEGIN  
...  
    INSERT INTO DEPARTMENTS ... ;  
...  
END;  
  
GRANT EXECUTE ON add_dept TO SUSAN;
```

Referencing Objects in Subprograms

- He/she does NOT need any privileges on the objects referenced by SQL statements within the subprogram
- The user (SUSAN) does not need INSERT (or any other privilege) on the DEPARTMENTS table

```
CREATE OR REPLACE PROCEDURE add_dept ...  
IS BEGIN  
...  
    INSERT INTO DEPARTMENTS ... ;  
...  
END ;  
  
GRANT EXECUTE ON add_dept TO SUSAN;
```

This can seem strange at first, but there is a good reason for it. If SUSAN needed (and was therefore GRANTED) INSERT privilege on DEPARTMENTS, she would be able to access the table's data without using the ADD_DEPT procedure. The Oracle privilege system means that users can access the data ONLY by invoking a carefully written and well-tested subprogram.

This means DBAs and programmers must be extra-careful about granting privileges on database objects that contain or access confidential data.

Privileges on Referenced Objects

- Someone must have privileges on the referenced objects
- Who is it?
- The subprogram owner (creator) must hold the appropriate privileges on the objects referenced by the subprogram – this is called Definer's Rights

```
(Table owner or DBA): GRANT INSERT ON departments TO TOM;  
(Tom) CREATE OR REPLACE PROCEDURE add_dept ...  
    IS BEGIN  
        ...  
        INSERT INTO DEPARTMENTS ... ;  
        ...  
    END;  
(Tom) GRANT EXECUTE ON add_dept TO SUSAN;
```

Definer's Rights is the reason why DBAs are often unwilling to allow application developers to create subprograms in production databases, because (since the privileges are checked every time the subprogram is invoked) the developer would need to hold the object privileges for the lifetime of the subprogram.

Privileges on Referenced Objects

- The owner's privileges are checked when the subprogram is created or replaced, and also every time the subprogram is invoked
- In this example, TOM creates a procedure that SUSAN needs to invoke:

```
(Table owner or DBA): GRANT INSERT ON departments TO TOM;  
(Tom) CREATE OR REPLACE PROCEDURE add_dept ...  
      IS BEGIN  
        ...  
        INSERT INTO DEPARTMENTS ... ;  
        ...  
      END;  
(Tom) GRANT EXECUTE ON add_dept TO SUSAN;
```


Privileges on Referenced Objects

- Below is another example
- BILL owns the STUDENTS and GRADES tables
- HANNAH needs to create a procedure that JIEP needs to invoke:

```
(Hannah)CREATE OR REPLACE PROCEDURE student_proc ...  
  IS BEGIN  
    SELECT ... FROM bill.students JOIN bill.grades ...;  
    UPDATE bill.students ...;  
    ...  
  END;  
  
(Jiep)  BEGIN    hannah.student_proc (...);  END;
```

- Who needs which privileges on which objects?

Answer: HANNAH needs SELECT and UPDATE on BILL.STUDENTS. HANNAH also needs SELECT on BILL.GRADES. JIEP only needs EXECUTE on HANNAH.STUDENT_PROC.

System Privileges

- On the previous slide, HANNAH created a procedure
- What privilege(s) did HANNAH need in order to do this?
- Yes, HANNAH needs suitable object privileges on BILL's tables
- She also needs the CREATE PROCEDURE system privilege:

```
(DBA) GRANT CREATE PROCEDURE TO hannah;
```

- Although the name of the privilege is CREATE PROCEDURE, it also allows HANNAH to create functions and packages

CREATE PROCEDURE allows the grantee to create any kind of PL/SQL subprogram, except a trigger, within their own schema. To create a trigger, you need the CREATE TRIGGER privilege (triggers will be discussed later in this course).

Terminology

- Key terms used in this lesson included:
 - ALTER privilege
 - EXECUTE privilege
 - INDEX privilege
 - Object privilege
 - REFERENCES privilege

- ALTER privilege – Allows the grantee to ALTER the table.
- EXECUTE privilege – Allows the user to invoke and execute a PL/SQL subprogram.
- INDEX privilege – Allows the grantee to create indexes on the table.
- Object privilege – Allows the use of a specific database object, such as a table, a view, or a PL/SQL procedure, by one or more database users.
- REFERENCES privilege – Allows the grantee to check for the existence of rows in a table using foreign key constraints.

Summary

- In this lesson, you should have learned how to:
 - List and explain several object privileges
 - Explain the function of the EXECUTE object privilege
 - Write SQL statements to grant and revoke object privileges

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